Test Method for Zipper Strength	Test Method #: VF- B17 Page 1 of 12
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PURPOSE AND SCOPE: To determine the strength of the zippers and zipper parts

1. Crosswise Strength - to determine crosswise strength which measures the resistance of a zipper to such failures as tape rupture, un-meshing, or element separation when the zipper is side stressed during use.
2. Element Pull-Off – to determine element strength which measures an element's ability to resist being fractured or pulled from the bead of the tape when the zipper is side stripped during use.
3. Element Slippage- to determine element attachment security and measures an element's ability to remain on the tape bead in proper position to adjacent elements so that the zipper operation can be maintained.
4. Top Stop Holding Strength– to determine top stop attachment strength which measures the ability of the top stop to prevent travel of the slider beyond the end of the chain.
5. Bottom Stop Holding Strength, Slider – to determine bottom stop attachment strength, which measures the ability of the bottom stop to resist failure caused by stress applied longitudinally to the bottom stop through the slider.
6. Bottom Stop Holding Strength, Crosswise - to determine bottom stop attachment strength, which measures the ability of the bottom stop to hold the two stringers of the chain together when the zipper is side stressed at the bottom stop
7. Bottom Stop Holding Strength, Stringer Separation - to determine bottom stop attachment strength, which measures the ability of the bottom stop to resist failure caused by such things as tape bead rapture, element separation from bead or bottom stop displacement due to stress applied through the stringers.
8. Bridge Top Stop Strength, Stringer Separation – to determine bridge top stop attachment strength, which measures the ability of the bridge top stop to remain in place holding the stringers of zippers together and limiting slider travel when the stop is stressed through the stringers.
9. Separable Pin, Pull-off – to determine separable pin attachment strength which measures the ability of the separable pin to resist displacement on the tape bead when a longitudinal force is applied.
10. Fixed Retainer, Pull-off – to determine fixed retainer attachment strength which measures the ability of the fixed retainer to resist displacement on the tape bead when a longitudinal force is applied.

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11. Separating Unit, Crosswise – to determine meshed separating unit attachment strength which measures the ability of the separating unit to resist failure due to separation from the tape bead, tape fracture or separation of the unit caused by lateral stressing of the zipper at the separating unit.

12. Resistance to Twist of Pull and slider – to determine slider pull strength which measures the ability of a slider pull or pull assembly, either or both, to resist deformation or rupture when a torsional stress is applied on the pull during operation of the zipper in the end – product.

13. Torsional Resistance to Pull-Off of slider Pull – to determine slider pull strength which measures the ability of a slider pull or pull assembly, either or both, to resist deformation or rupture when torsional stress is applied on the pull during operation of the zipper in the end product.

14. Resistance to Pull-off of Slider Pull – to determine slider pull strength, which measures the resistance of the pull or pull assembly, either or both, to rupture, deformation or separation when a stress is perpendicularly applied through the pull during operation of the zipper in the end-product.

15. Resistance to Angular Pull –Off of Slider Pull – to determine slider pull strength which measures the resistance of the pull or pull assembly, either or both, to rupture, deformation or separation when a stress is applied at an angle during operation of the zipper in end – product.

16. Holding Strength of Slider Lock – to determine the locking strength of a slider which measures the ability of the locking mechanism of a slider to hold the slider in a locked position on the chain when stress is applied through the stringers.

REFERENCES: D 2061-07

EQUIPMENTAND MATERIAL:

1. Tensile testing machine equipped with chart recorder
2. Clamping jaws, with small hook attachment or fixture assembly
3. Complete assembled zippers with sliders and lock, a chain only, or a component.

SAMPLE PREPARATION:

1. Condition the test sample in a standard controlled environment @ 23.0±2.0 °C temperature and at 55±10 % RH for at least 24 hours.

2. Prepare three (3) zippers per test method. The grips and fixtures should match the type of test to be conducted.

REPETITIONS: Test three (3) fasteners /zippers for each of the tests

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TEST PROCEDURE:

Test should be conducted at standard laboratory environment @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

I. Strength of Chains and Elements:

1. Crosswise Strength:

   Secure the tapes of the zipper or chain in the clamps of the tensile testing machine with the edges of the jaws parallel to the chain and approximately 3 mm from the outer edge of the inter-lockable elements or the outer edge of the beads if the bead extends beyond the elements. The ends of the front jaws shall be at least 25 mm from the slider, stops or end of the of the meshed chain. Apply an increasing load until the elements pull off the bead, until the tape separates, or until failure of some other kind occurs. Record the nature of failure and the breaking load to the nearest 0.1 kgf.

2. Element Pull-Off:

   Secure the element pull-off fixture (figure1) in the top clamp of the testing machine. Then adjust the fixture to grasp a single element on a stringer. Secure the stringer in the bottom clamp of the testing machine as described in fig 1.1. Apply an increasing load until the element is pulled off or other failure occurs. Record the nature of the failure and the load at failure to the nearest 0.1 kgf.

3. Element Slippage, Lengthwise:

   The test specimen shall be cut to approximately 127 mm in length. Separate the chain into stringers and hold a stinger so the elements are positioned as in figure 2. Remove the second element from the top of the stringer, taking care not to damage the bead. Cut through the bead and the tape in the width direction below the first element and above the remove element. The tape and the bead may be cut either partially completely across as long as the element to be tested is not hindered inn its movement along the bead. Cut the tape the sharp scissors, cutters or nippers using a single stroke, ignoring blooming if it's occurs. The element to be tested is the one just below the element that was removed. Secure the element slippage fixture (figure 2) in the top clamp and position the two prongs at the end of the short leg of the "L" in the horizontal alignment underneath the element. Secure the stringer in the lower clamp of the testing machine in vertical alignment, positioning the clamps approximately 76 mm below the "L" of the fixture in the upper clamp. Apply an increasing load until the element slips off the end of the cut bead or until the element slips off the end of the cut bead or until other failure occurs. Record the nature of the failure to the nearest 0.1kgf.

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II. Holding Strengths of Stops:

1. Top Stop Holding Strength:

   Secure the fixture in the upper clamp of the testing machine and hook the pull of the slider on the fixture (fig 3). Position the slider body at the point where normally checked in its free movement by the stop or stops to be tested. Secure the zipper in the lower clamp of the testing machine as illustrated in fig 4. The distance between the top edge of the lower clamp and mouth of the slider shall be approximately 76 mm. Apply an increasing load until the stop or stops pull off, until the tape breaks, or until failure of some other occurs. Record the nature of the failure and the load at failure to the nearest 0.1 kgf.

2. Bottom Stop Holding Strength, Slider: Secure the fixture in the upper clamp of the testing machine and hook the pull of the slider on the fixture (fig 3). Position the slider at the point where normally checked in its free movement by the bottom stop to be tested. Secure the two stringers in the lower clamp of the testing machine as illustrated in figure 5, taking care to equalize the lengths of the two stringers between the clamps. The angle included between the stringers shall be such as to prevent the catching of the elements on the flanges or on the diamond. The distance between the top edge of the lower clamp the nearest surface of the slider body shall be approximately 76mm. Apply an increasing load until the stop pulls off, or until the tape breaks, or until failure of some other kind occurs. Record the nature of failure and the load of failure to the nearest 0.1kgf.

3. Bottom stop Holding Strength, Crosswise:

   Remove the slider from the zipper. Open the chain by pulling the stringer apart all the way to the bottom stop. From both stringers remove the elements adjacent to but under the bottom stop for a distance of approximately 13mm. Secure the tapes of the zipper in the clamps of the testing machine with the edges of the jaws parallel to and approximately 3 mm from the sides of the bottom stop, which shall be centrally located in the clamps as shown in fig 6. Apply the load until the stop pull apart, until the tape breaks, or until failure of some kind occurs. Record the nature of the failure and the load at the failure to the nearest 0.1kgf.

4. Bottom stop Holding Strength, Stringer Separation:

   Position the slider body so that its mouth is against the bottom of the stop to be tested. In the case of the entering type bottom stop, position the slider at the point where normally checked in its free movement. Set the opposing clamps of the testing machine approximately 76mm apart and secure one of the stringers in the upper clamp and the other in the lower clamp of the tensile testing machine approximately 76mm apart. Secure the stringers in the upper and lower clamps of the tensile testing machine with the slider body positioned along the axis of the

clamps and midway between them as shown in fig 7. Apply an increasing load until the stop pulls off, until the tape breaks, or until failure of some other kind occurs. Record the nature of the failure and the load at the failure to the nearest 0.1kgf.

5. Bridge Top Stop Strength, Stringer Separation:

   With the opposing clamps of the tensile testing machine set approximately 76 mm apart, separate the two stringers and secure them in the opposing clamps of the tensile machine. Position the stop along the axis of the clamps and midway between them as shown in fig 8. Apply an increasing load until the stop pulls off, until the tape breaks, or until failure of some other kind occurs. Record the nature of the failure and the load of failure to the nearest 0.1kgf.

III. Holding Strengths of Separable Units:

1. Separable Pin, Pull – Off:

   Separate the two stringers and, adjacent to the separable pin, remove one or more elements from the stringer. Secure the separable pin fixture (fig 9) in the upper clamp of the testing machine. Then seat the separable pin on the fixture notch as shown in fig 9. With a distance of approximately 76 mm between the seated end of the pin and the top of the lower clamp, secure the stringer in the lower clamp. Apply an increasing load until the separable pin pulls- off, until the tape breaks, or until failure of some other kind occurs. Record the nature of the failure and the load of failure to the nearest 0.1kgf.

2. Fixed Retainer ,Pull –Off:

   Secure the fixture (fig 10) in the upper clamp of the testing machine. Separate the two stringers and position the fixed retainer on the upper edges of the slot of the fixture and, with an approximate 76 mm distance between the lower edge of the fixed retainer and the upper edge of the lower clamp, secure the stringer in the lower clamp as shown in fig 11. Apply an increasing load until the retainer pulls off, until tape breaks, or until failure of some other kind occurs. Record the nature of the failure and the load of failure to the nearest 0.1kgf.

3. Separating Unit, Crosswise :

   Secure the tape of the zipper in the clamps of the testing machine as shown in fig 12 , with the edges of the jaws parallel to ,and approximately 3mm from the sides of the separating unit , whether the latter is of fixed or movable type . Position the separating unit so that the exposed end of the separable pin is aligned with the sides of the front jaws as shown in fig 12. Apply the load until the separating unit comes apart, until the tape breaks, or until failure of some other kind occurs. Record the nature of the failure and the load of failure to the nearest 0.1kgf

IV. Resistance to Twist of Pull and Slider:

1 Attach the slider in the pull twist test fixture. Position the pull in the clamp , aligning the front surface of the clamp with the midpoint of the pull as shown in fig 18 .Turn the testing drum clockwise until the clearance between the lug and the pull trunnions is taken up , and then set the dial to zero.

Use the appropriate torque wrench and at the rate of approximately 9% apply the torsional force required by the applicable specification. Release the torsional force. Turn the testing drum by hand in the direction of the torsional force until the clearance is taken up. Read the dial and record the amount of permanent twist. Remove the slider.

On the second specimen, repeat the test as directed in IV.1 and IV.2, except that the direction of rotation shall be counterclockwise.

Examine the specimens and note the effects of the test, such as breaking or deformation of the lug or any other part of the assembly.

V. Torsional Resistance of Slider Pull for Removal of Components:

Attach the slider body to the selected pull twist test fixture. Position the pull in the clamp aligning the front surface of the clamp with the midpoint of the pull as shown in fig 18 View “A”.

Use the appropriate loading devise. Start to turn the loading device in clockwise direction Applying force evenly so that within 5 s (36 ° /s) one of the following is achieved :

- The torque wrench has traveled 180 degrees from it’s original position or
- The required force has been released or
- Breakage or disassembly of the pull below the required force is observed (should breakage of the pull occur no further testing is required).
- The maximum rotation or required torque should be maintained for 10 s.

After 10s has elapsed release the torsional force by turning the torque devise in the opposite counterclockwise direction until the load device reads zero force and a relaxed, tensionless state is achieved. Continue turning the torque device in the counterclockwise direction until a load begins to be applied. The position of the pull at this point be the starting point for the second test in the counter clockwise direction.

Repeat the procedure but in the counterclockwise direction on the same specimen.

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Test to failure: should the pull reach the required force after the 10 s has elapse in the counterclockwise direction the pull may be tested to failure by continuing to turn the loading device in the counterclockwise direction until breakage occurs. The ultimate breaking force may be recorded.

Examine the specimen and note the effects of the test, such as breaking or deformation of any part of assembly.

VI. Resistance to Pull – Off slider Pull:

Secure the slider pull fixture in the upper clamp of the testing machine. Place the slider body horizontally on the fixture, with the pull projecting vertically downward, so that both the bail and pull are clear and unrestricted as shown in fig 19. Position on the trunnions midway in the length of the bails which are of the elongated type. Hook the other part of the fixture into the pull, and secure it in the bottom clamp of the testing machine. Apply the load required by the applicable specification.

Examine the specimen and note the effect of the test, such as breakage, rupture, deformation or separation of the pull from the slider body of the zipper.

VII. Resistance to Angular Pull-Off Of Slider Pull:

Secure the angular pull fixture in the upper clamp of the testing machine. Place the slider body in the fixture, with the pull projecting vertically downward, so that both the bail and the pull are clear and unrestricted. This is accomplished by abutting the adjusting screw against the diamond end of the slider body. See fig 20. Hook the other part of the fixture into the pull and then secure it in the bottom clamp of the testing machine. Apply the load required by the applicable specification.

Examine the specimen and note the effects of the test, such as breakage, rupture, deformation , or separation of the pull from the slider body.

VII. Holding Strength of Slider Lock:

Set the distance between the clamps of the testing machine at approximately 76 mm. Locate and lock the slider midway between the ends of the chain. Secure one of the stringers emerging from the throat of the slider in the upper clamp and one in the lower clamp of the testing machine.

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Position the slider body along the axis of the clamps and midway between them as shown in Fig 21. Apply the load required by the applicable specification.

Examine the specimen to determine the effects of the test, such as lock slippage, lock damage, tape or inter-lockable element damage affecting zipper functioning.

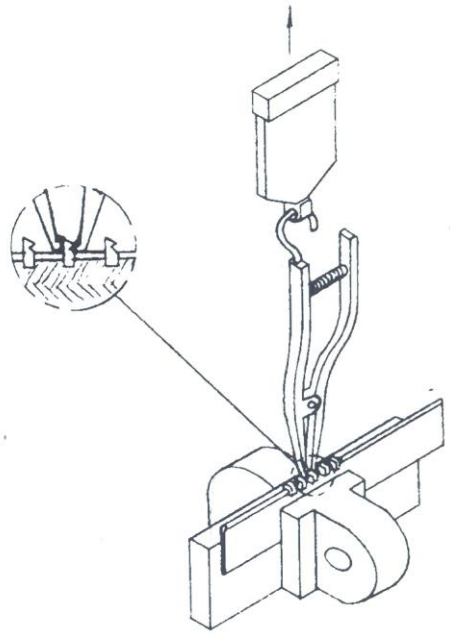


FIG. 1 Fixture for Element Pull-Off Test

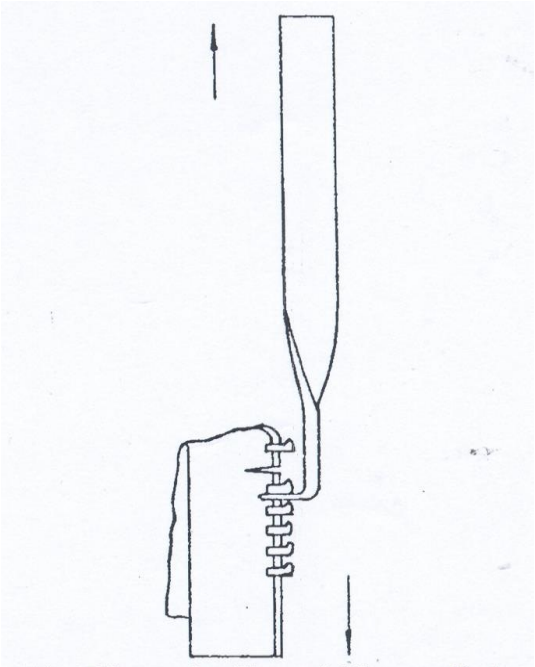


FIG. 2 Fixture for Element Slippage Test

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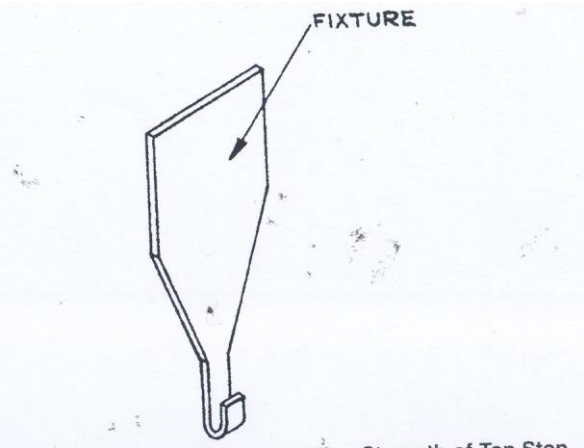


FIG. 3 Fixture for Test for Holding Strength of Top Stop

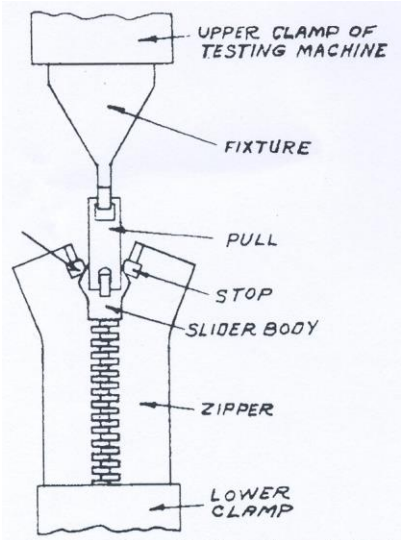


FIG. 4 Top Stop Holding Strength Test

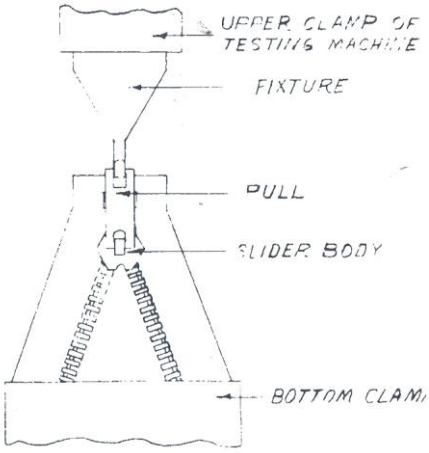


FIG. 5 Bottom Stop Holding Strength Test

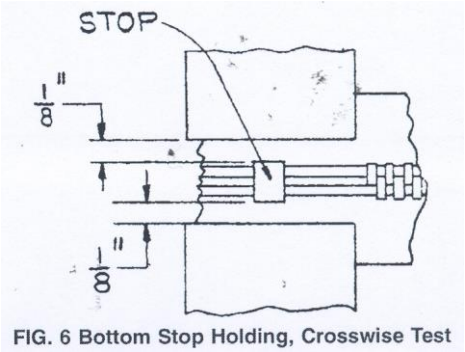


FIG. 6 Bottom Stop Holding, Crosswise Test

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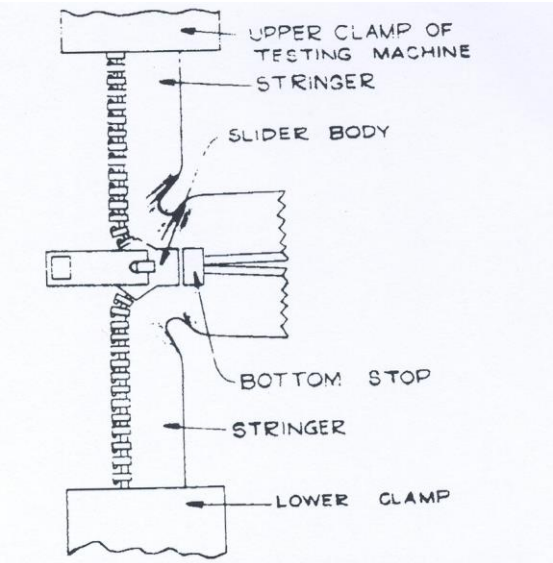


FIG. 7 Bottom Stop Holding, Stringer Separation Test

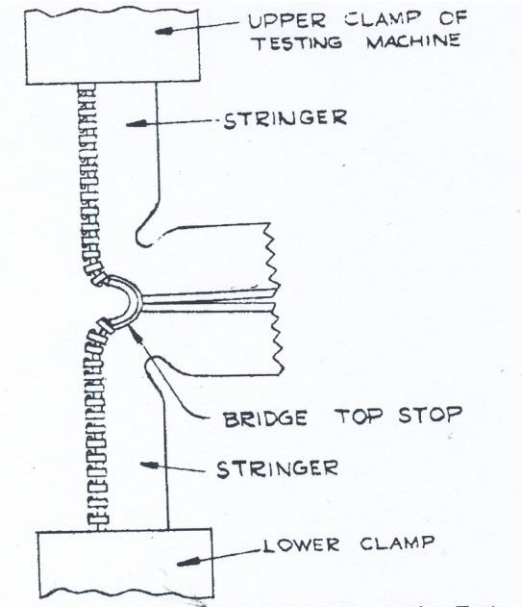


FIG. 8 Bridge Top Stop, Stringer Separation Test

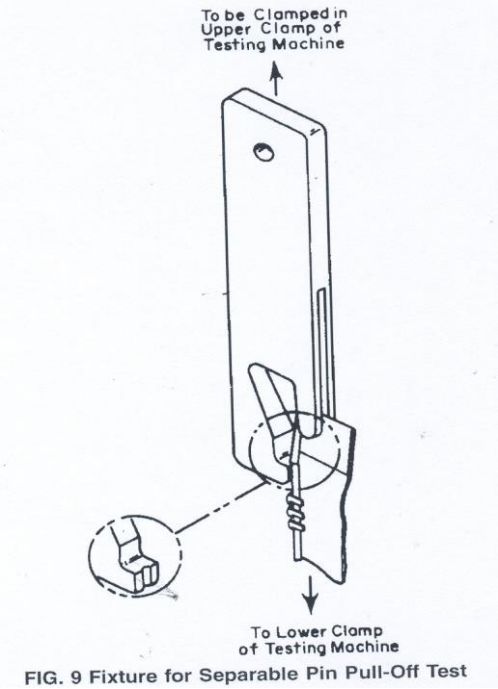
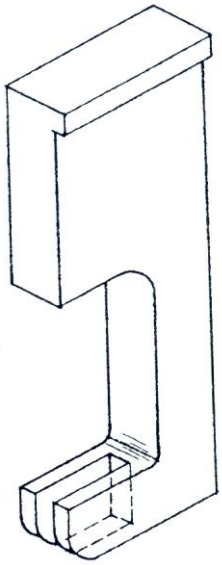


FIG. 9 Fixture for Separable Pin Pull-Off Test


FIG. 10 Fixture for Retainer Pull-Off Test

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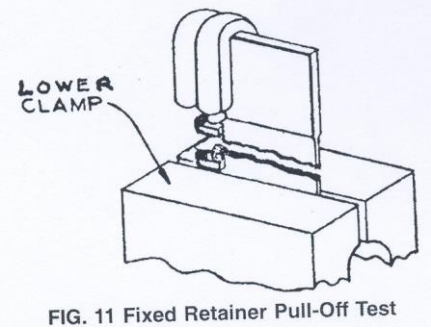


FIG. 11 Fixed Retainer Pull-Off Test

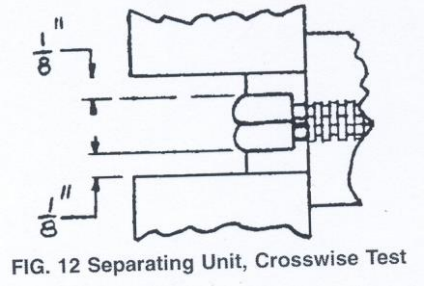


FIG. 12 Separating Unit, Crosswise Test

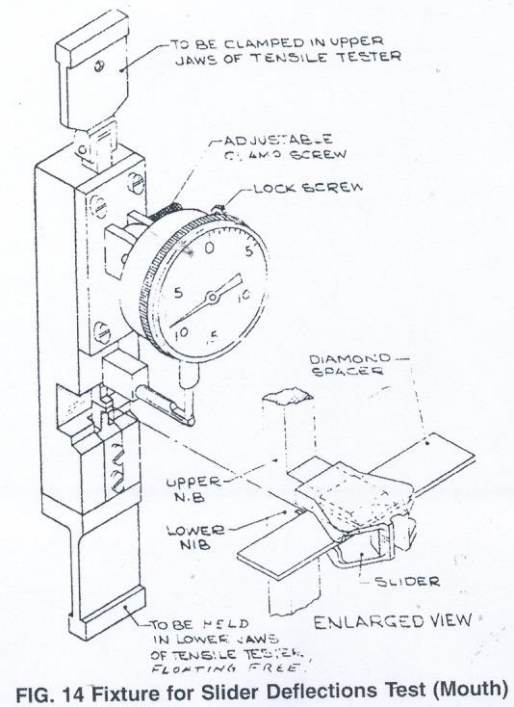
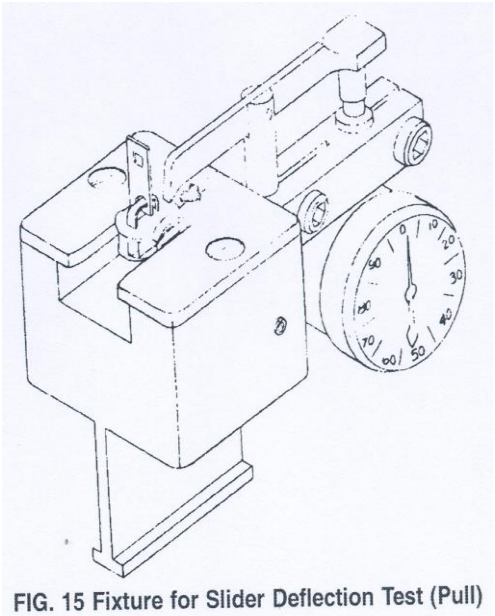


FIG. 14 Fixture for Slider Deflections Test (Mouth)


FIG. 15 Fixture for Slider Deflection Test (Pull)

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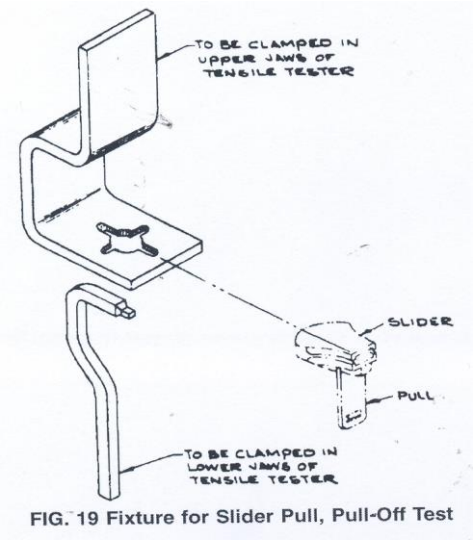


FIG. 19 Fixture for Slider Pull, Pull-Off Test

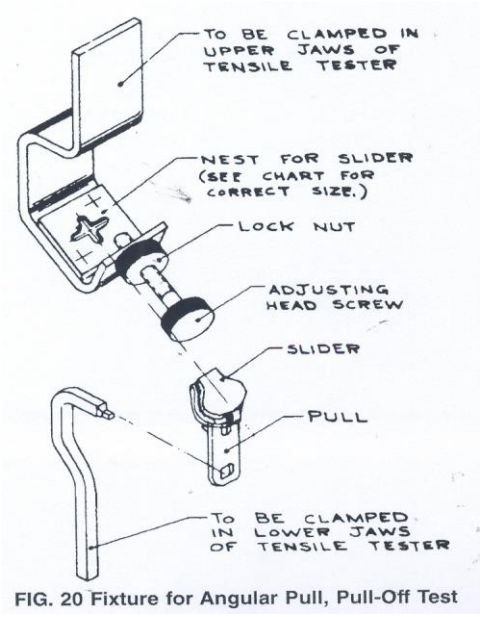


FIG. 20 Fixture for Angular Pull, Pull-Off Test

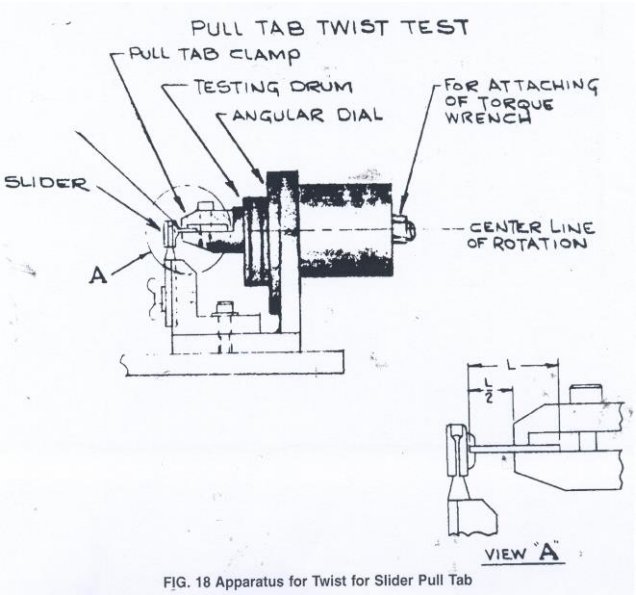


FIG. 18 Apparatus for Twist for Slider Pull Tab

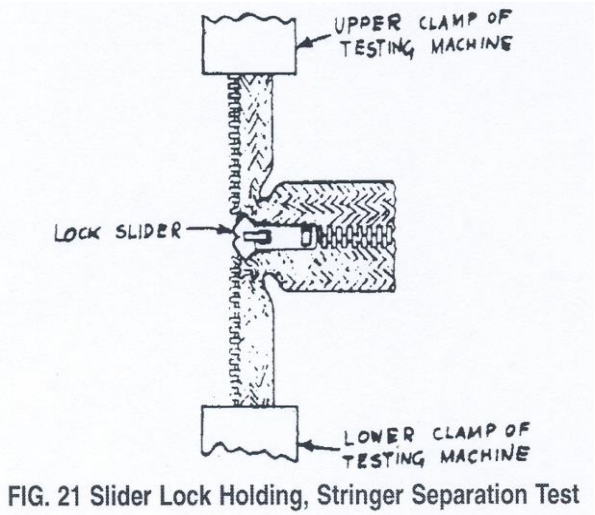


FIG. 21 Slider Lock Holding, Stringer Separation Test